

EXCITATION-DEPENDENT IDENTIFIABILITY OF LATENT HIGHER-ORDER STRUCTURE FROM EDGE FLOWS

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ABSTRACT

Which triangles of a network carry higher-order interactions? Edge flows reveal the latent filled-triangle set $\mathcal{S}$ only through their curl, and we show that *what* is identifiable is governed by the excitation covariance $\Gamma_{\mathcal{S}}$ of the triangle signals. Our main theorem is a trichotomy: for positive-diagonal $\Gamma_{\mathcal{S}}$ (known or unknown), the weighted support is identifiable at *any* rank deficiency of B_2 , because the lifted signatures $\{b_r b_r^\top\}$ are always linearly independent; for unknown arbitrary PSD $\Gamma_{\mathcal{S}}$, the achievable covariance family determines exactly the curl subspace $\text{im } B_{2,\mathcal{S}}$ and nothing finer; and at the projector excitation $\Gamma_{\mathcal{S}} = \sigma_c^2 (B_{2,\mathcal{S}}^\top B_{2,\mathcal{S}})^+$ the covariance collapses to $\sigma_n^2 I + \sigma_c^2 P_{\text{im}}$ exactly, so equal-image supports are indistinguishable at every SNR and sample size (noise level known throughout). In the diagonal class we give a lifted-covariance non-negative least-squares estimator that is consistent with a fully explicit $O(1/N)$ failure bound, validated on K_4 – K_8 against subspace and greedy baselines, including an excitation interpolation under which equal-image distinguishability provably vanishes. On real data, the same curl statistic *localizes* genuine unplanted rotational structure — tropical cyclones in ERA5 winds — against a same-field vorticity reference and the independent IBTrACS archive. Code, data, and a 46-test suite certifying every constant are released.

Index Terms— Topological signal processing, simplicial complexes, identifiability, Hodge decomposition, covariance estimation, edge flows.

1. INTRODUCTION

Edge flows — currency exchange, traffic, ranking comparisons — carry higher-order structure on the triangles of a simplicial complex [1, 2, 3]. Topological signal processing builds filters and detectors on the Hodge Laplacian [4, 5, 6, 7, 8], and a recurring primitive is that the set $\mathcal{S}$ of “filled” triangles is unknown. One line of work *estimates* $\mathcal{S}$ algorithmically: MAP estimation [9], greedy topology learning [10], sparse cell-complex augmentation [11]. Another *detects* whether a signal lies in a known Hodge subspace: matched topological subspace detectors [12] test deviation from, e.g., curl-free structure with the subspace given a priori. Between the two sits the question this paper answers: *when is the latent structure that generates the subspace identifiable at all — and what part of it?*

The answer, it turns out, is not a property of the geometry alone: it depends on the *excitation*. We model triangle signals as $y_{\mathcal{S}} \sim \mathcal{N}(\mathbf{0}, \Gamma_{\mathcal{S}})$ with PSD excitation covariance $\Gamma_{\mathcal{S}}$ and show that the identifiable object moves with the class of $\Gamma_{\mathcal{S}}$ (Theorem 1): from the full weighted support (positive diagonal, any rank deficiency), through exactly the curl subspace $\text{im } B_{2,\mathcal{S}}$ (unknown arbitrary

PSD), to provable indistinguishability of equal-image supports (projector excitation). This corrects a folklore “rank obstruction” — and also bounds what image-level methods such as subspace detectors [12] can ever resolve. Our tools are classical — detection theory and Fano arguments [13, 14], in the lineage of support-recovery and sparse-covariance limits [15, 16, 17, 18] — applied through a reduction (curl annihilation, after HodgeRank’s curl-as-inconsistency reading [19]) that makes them bite. Contributions:

1. *Theory*: the excitation trichotomy (Thm. 1), with global analytic separation constants for the isotropic class — equal-image binary supports are separated by at least 9 (attained by adding a tetrahedron’s fourth face) and equal-cardinality supports by 16 (swaps), proved for *every* clique complex via a Johnson-graph eigenvalue argument — and Chernoff exponents, for fixed graph and support as $\rho_2 \rightarrow 0$, whose worst case equals one isolated-triangle detection (Thm. 2).
2. *Achievability*: a lifted-covariance NNLS estimator for the diagonal class, consistent with a fully explicit $O(1/N)$ failure bound in which every constant is derived and test-certified (Thm. 3); Monte-Carlo validation on K_4 – K_8 with random supports against oracle-aided subspace and greedy baselines, and an excitation-interpolation experiment in which equal-image distinguishability collapses as predicted (Fig. 1).
3. *Real data, carefully positioned*: curl-based *vortex localization* — not topology recovery — of unplanted tropical-cyclone circulation in ERA5 winds, validated against a same-field vorticity reference and the independent IBTrACS archive, with a classical baseline and window-bootstrap confidence intervals (Fig. 3).

2. MODEL AND CURL REDUCTION

Fix a graph $(\mathcal{V}, \mathcal{E})$ with node–edge incidence B_1 and, for a candidate triangle set $\mathcal{T}$ (3-cliques), edge–triangle incidence $B_2 \in \mathbb{R}^{|\mathcal{E}| \times p}$; $B_1 B_2 = \mathbf{0}$, and each column b_r has three ± 1 entries, so $G = B_2^\top B_2$ has $G_{\tau\tau} = 3$ and $G_{\sigma\tau} \in \{0, \pm 1\}$ (nonzero iff the triangles share an edge). We observe N i.i.d. snapshots

$$f_t = B_1^\top a_t + B_{2,\mathcal{S}} y_t + h_t + n_t, \quad t = 1, \dots, N, \quad (1)$$

with gradient nuisance $a_t \sim \mathcal{N}(\mathbf{0}, \sigma_g^2 I)$, harmonic nuisance h_t , noise $n_t \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 I)$, and triangle excitation

$$y_t \sim \mathcal{N}(\mathbf{0}, \Gamma_{\mathcal{S}}), \quad \Gamma_{\mathcal{S}} \succeq \mathbf{0}, \quad (2)$$

generalizing the isotropic $\Gamma_{\mathcal{S}} = \sigma_c^2 I$ that algorithmic formulations implicitly adopt (cf. [9]); making $\Gamma_{\mathcal{S}}$ explicit is the point of this paper. Curl annihilation ($B_2^\top B_1 = \mathbf{0}$, $B_2^\top h = \mathbf{0}$) [19] removes

both nuisances exactly; in an orthonormal basis $\mathbf{Q}$ of $\text{im } \mathbf{B}_2$ ($r = \text{rank } \mathbf{B}_2$), the sufficient statistic $\mathbf{z}_t = \mathbf{Q}^\top \mathbf{f}_t$ is Gaussian with

$$\Sigma_z(\mathcal{S}, \Gamma_S) = \sigma_n^2 \mathbf{I}_r + \mathbf{U}_S \Gamma_S \mathbf{U}_S^\top, \quad \mathbf{U} = \mathbf{Q}^\top \mathbf{B}_2, \quad (3)$$

with $\|\mathbf{u}_\tau\|^2 = 3$ and $\mathbf{u}_\sigma^\top \mathbf{u}_\tau = G_{\sigma\tau}$. Identifiability of $\mathcal{S}$ from (3) is the object of study; for the isolated-triangle scalar test this reduces to the classical two-variance problem with curl-SNR $\rho = 3\sigma_c^2/\sigma_n^2$ [13] (supplement §S1–S3 develops that line: detection thresholds, Fano converses, decorrelated detectors, plug-in estimation).

3. EXCITATION-DEPENDENT IDENTIFIABILITY

Lemma 1 (Lifted spark). *For any candidate set of distinct 3-cliques of a simple graph, the matrices $\{\mathbf{b}_\tau \mathbf{b}_\tau^\top\}_{\tau \in \mathcal{T}}$ are linearly independent.*

Proof. A pair of distinct edges lies in at most one common triangle, so for $e \neq e'$ both in τ , the (e, e') entry of $\sum_\sigma w_\sigma \mathbf{b}_\sigma \mathbf{b}_\sigma^\top$ equals $\pm w_\tau$: every coefficient is read off a distinct matrix entry. $\square$

Theorem 1 (Excitation trichotomy). *Let σ_n^2 be known and write $\mathbf{M}(\mathcal{S}, \Gamma) = \mathbf{U}_S \Gamma \mathbf{U}_S^\top$ in (3). (a) Positive diagonal. If $\Gamma_S = \text{diag}(\gamma_\tau)_{\tau \in \mathcal{S}}$ with $\gamma_\tau > 0$ (known or unknown), then $\mathbf{M} = \sum_{\tau \in \mathcal{S}} \gamma_\tau \mathbf{u}_\tau \mathbf{u}_\tau^\top$ and, by Lemma 1, the map $(\mathcal{S}, \Gamma_S) \mapsto \Sigma_z$ is injective: the weighted support is identifiable at any rank deficiency of $\mathbf{B}_2$, including all $\binom{n}{3}$ triangles of K_n . (b) Arbitrary PSD. If Γ_S is unknown and unrestricted PSD, then $\{\mathbf{M}(\mathcal{S}, \Gamma) : \Gamma \succeq 0\} = \{\mathbf{M} \succeq 0 : \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_S\}$: the achievable covariance family determines, and is determined by, $\text{im } \mathbf{B}_{2,S}$ — so supports are distinguishable as composite hypotheses at most up to equal curl image, and nothing finer than the image is identifiable (pointwise, the image itself is identified when $\Gamma_S \succ 0$; singular excitations can conceal part of it). (c) Projector excitation. For $\Gamma_S = \sigma_c^2 (\mathbf{B}_{2,S}^\top \mathbf{B}_{2,S})^+$, $\mathbf{M} = \sigma_c^2 \mathbf{P}_{\text{im } \mathbf{U}_S}$ exactly; supports with equal curl image induce identical snapshot distributions and are indistinguishable at every SNR and every N .*

Proof sketch. (a) Read the weights off the lifted representation (spark). (b) $\subseteq$ is immediate; $\supseteq$: given PSD $\mathbf{M}$ with $\text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_S$, take $\Gamma = \mathbf{U}_S^\top \mathbf{M} (\mathbf{U}_S^\top)^\top \succeq 0$. (c) $\mathbf{A}(\mathbf{A}^\top \mathbf{A})^+ \mathbf{A}^\top = \mathbf{P}_{\text{im } \mathbf{A}}$ (SVD). On K_n , $\dim \text{im } \mathbf{B}_2 = \binom{n-1}{2}$ (a $3/n$ DoF ratio), and equal-image supports exist whenever a tetrahedron is hosted ($\mathbf{b}_{012} - \mathbf{b}_{013} + \mathbf{b}_{023} - \mathbf{b}_{123} = \mathbf{0}$). Full proofs and numerical certificates: supplement §S4, tests/test_excitation.py. $\square$

Remark 1 (Scope). (i) Deterministic unknown signals are the degenerate boundary of class (b): identifiability only modulo equal image. (ii) On K_n with σ_n unknown, class (a) retains exactly one ambiguous direction, since $\sum_{\tau \in \mathcal{T}} \mathbf{u}_\tau \mathbf{u}_\tau^\top = n \mathbf{I}_r$: a uniform weight shift trades off against the noise floor. (iii) Class (b) delimits image-level methods — e.g. matched subspace detectors [12] — whose population statistics see only $\text{im } \mathbf{B}_{2,S}$: without excitation structure, no method can do better.

Theorem 2 (The price in the isotropic class). *Let $\Gamma_S = \sigma_c^2 \mathbf{I}$, $\rho_2 = \sigma_c^2/\sigma_n^2$. For any clique complex: distinct binary supports with equal curl image satisfy $\|\mathbf{M}_S - \mathbf{M}_{S'}\|_F^2 \geq 9\sigma_c^4$, with equality exactly for single-face differences (e.g. $S' = S \cup \{\text{fourth face}\}$), and $\geq 16\sigma_c^4$ under equal cardinality (tetrahedral swaps). For fixed graph and support pair, as $\rho_2 \rightarrow 0$ the optimal-test Chernoff exponents are $\frac{9}{16}\rho_2^2(1+o(1))$ and $\rho_2^2(1+o(1))$ respectively — and $\frac{9}{16}\rho_2^2$ equals the isolated-triangle detection exponent $C(\rho) = \rho^2/16$ at $\rho = 3\rho_2$: the*

hardest equal-image decision costs one isolated-triangle detection, so rank deficiency is free at the exponent level and only the Fano log-multiplicity of the $4\binom{n}{4}$ tetrad hypotheses ($\asymp \log n$ on K_n) reflects the geometry.

Proof sketch. Separations: $\|\sum_\tau c_\tau \mathbf{b}_\tau \mathbf{b}_\tau^\top\|_F^2 = \mathbf{c}^\top (9\mathbf{I} + \mathbf{A}) \mathbf{c}$ for ternary $\mathbf{c}$, where $\mathbf{A}$ is the share-an-edge adjacency — an induced subgraph of the Johnson graph $J(n, 3)$, so $\lambda_{\min}(\mathbf{A}) \geq -3$ by interlacing and $\text{sep} \geq 6\|\mathbf{c}\|^2$; enumerate $\|\mathbf{c}\|^2 \in \{1, 2\}$. Exponents: Bhattacharyya expansion $C_G = \frac{1}{16}\rho_2^2 \|\Delta \mathbf{M}\|_F^2 (1 + o(1))$. Test-certified: swap C_G/ρ_2^2 within 6×10^{-4} of 1 and subset matching the single-triangle exponent to 3×10^{-5} relative (at $\rho_2 = 10^{-4}, 10^{-5}$); separations exhaustive over all 2^{10} supports of K_5 . Proofs: supplement §S4. $\square$

Proposition 1 (Edge-sharing marginal laws). *Assume $\mathbf{B}_2$ full column rank and $\Gamma_S = \sigma_c^2 \mathbf{I}$. The GLS/BBLUE inversion $\hat{\mathbf{y}}_t = \mathbf{G}^{-1} \mathbf{B}_2^\top \mathbf{f}_t$ [20] satisfies, exactly, $\hat{\mathbf{y}}_t = \tilde{\mathbf{y}}_t + \mathbf{e}_t$ with $\mathbf{e}_t \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{G}^{-1})$: each coordinate is a two-variance test with effective SNR $\rho_\tau^{\text{eff}} = \sigma_c^2/(\sigma_n^2 (\mathbf{G}^{-1})_{\tau\tau})$ — the classical variance-inflation factor as the price of edge-sharing — and exact recovery admits a correlation-robust union-bound guarantee. The noise coordinates stay correlated ($\sigma_n^2 \mathbf{G}^{-1}$ is not diagonal), so the joint law does not factorize; benchmarks, the naive-detector failure mode, and proofs are in supplement §S1, S3.*

Proposition 2 (Detection thresholds through the reduction). *In the isotropic class, deciding one isolated triangle is the two-variance Gaussian test with variances $3\sigma_n^2$ vs. $9\sigma_c^2 + 3\sigma_n^2$ [13]: attaining error $\leq \delta$ at the exponent level requires the curl-SNR scale $\rho \geq \rho^*(N) = \Theta(\sqrt{\log(1/\delta)/N})$ ($C(\rho) = \rho^2/16 + o(\rho^2)$ [14]). For pairwise edge-disjoint candidates the curl coordinates are independent and exact recovery obeys the exact finite-sample product law $P(\hat{\mathcal{S}} = \mathcal{S}) = \prod_{\tau \in \mathcal{S}} (1 - P_{\text{miss}}) \prod_{\tau \notin \mathcal{S}} (1 - P_{\text{fa}})$ with χ_N^2 tails at the Bayes threshold — its 50% contour matches the empirical boundary to a median ratio 1.01 (Fig. 2). Recovering a uniformly random size- k support among p such candidates with error $\leq \epsilon$ requires $N \geq [(1-\epsilon) \log \binom{p}{k} - \log 2]/(k \text{KL}_1)$, $\text{KL}_1 = (\rho - \log(1+\rho))/2$ (Fano): joint recovery pays a $\log \binom{p}{k}$ factor. Signal-agnostic and sub-Gaussian variants, partial edge observation (q^{3k} laws), and plug-in calibration are developed and validated in supplement §S1–S3; these scalings are classical [15] — what is problem-specific is the reduction that produces ρ and the constants (3, 9).*

4. ACHIEVABILITY: LIFTED-COVARIANCE NNLS

In class (a) the theory says the weighted support is identifiable; the matching estimator is non-negative least squares on the lifted dictionary $\mathbf{A} = [\text{vec}(\mathbf{u}_\tau \mathbf{u}_\tau^\top)]_\tau$:

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \geq 0} \left\| \hat{\Sigma}_z - \sigma_n^2 \mathbf{I} - \sum_\tau w_\tau \mathbf{u}_\tau \mathbf{u}_\tau^\top \right\|_F^2, \quad (4)$$

with $\hat{\Sigma}_z = \frac{1}{N} \sum_t \mathbf{z}_t \mathbf{z}_t^\top$, $\hat{\mathcal{S}} = \{\tau : \hat{w}_\tau > w_{\min}/2\}$, and $w_{\min} = \min_{\tau \in \mathcal{S}} w_\tau$.

Theorem 3 (Consistency and explicit failure bound). *$\hat{\mathbf{w}} \rightarrow \mathbf{w}$ almost surely, and for every N ,*

$$\Pr[\hat{\mathcal{S}} \neq \mathcal{S}] \leq \frac{16[(\text{tr } \Sigma_z)^2 + \|\Sigma_z\|_F^2]}{N \sigma_{\min}^2(\mathbf{A}) w_{\min}^2}. \quad (5)$$

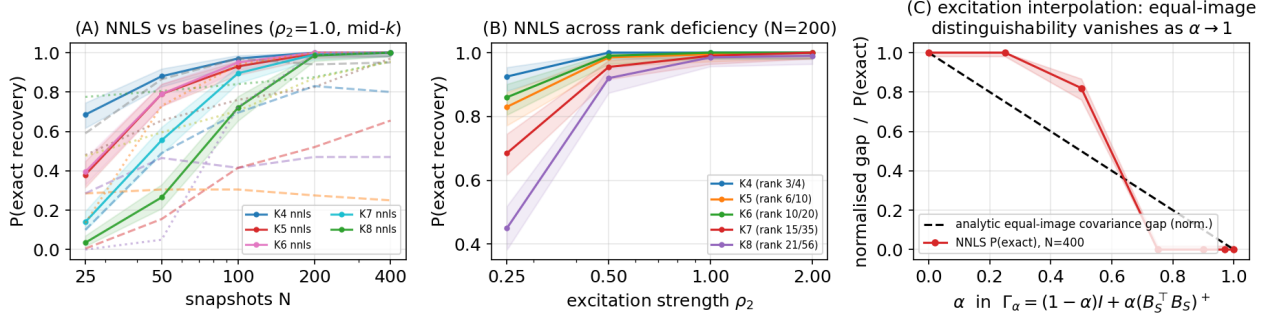


Fig. 1. Second-order achievability. (A) Exact recovery vs. N on K_4 – K_8 (random supports, $\rho_2 = 1$): NNLS (solid, 95% Wilson bands) vs. oracle-aided subspace (dashed) and greedy (dotted) baselines. (B) NNLS vs. ρ_2 at $N = 200$ across rank deficiency (legend: rank B_2/p). (C) Excitation interpolation on the K_5 tetrad: analytic equal-image gap (normalized) and NNLS recovery collapse together as $\alpha \rightarrow 1$ (Thm. 1(c)).

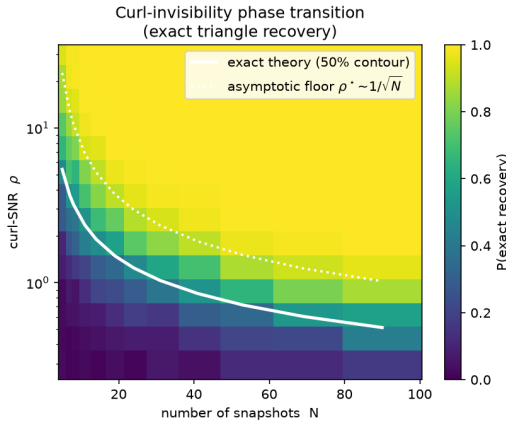


Fig. 2. Recovery threshold in the isotropic class (edge-disjoint planted complex, gradient/harmonic nuisance present). Colour: empirical $P(\text{exact recovery})$; solid: the exact finite-sample product law of Prop. 2; dotted: the Bonferroni-corrected Chernoff floor ($\delta \rightarrow \delta/p$), visibly the $1/\sqrt{N}$ law.

Proof sketch. Three derived steps, each test-certified: (i) cone-constrained least squares obeys the deterministic bound $\|\hat{\mathbf{w}} - \mathbf{w}\|_2 \leq 2\|\hat{\Sigma}_z - \Sigma_z\|_F / \sigma_{\min}(\mathbf{A})$ — optimality of $\hat{\mathbf{w}}$, feasibility of $\mathbf{w}$, Cauchy–Schwarz — with $\sigma_{\min}(\mathbf{A}) > 0$ by Lemma 1; (ii) for Gaussian snapshots $\mathbb{E}\|\hat{\Sigma}_z - \Sigma_z\|_F^2 = [(\text{tr } \Sigma_z)^2 + \|\Sigma_z\|_F^2]/N$ exactly (Wishart second moments); (iii) Markov at threshold $w_{\min}/2$. The bound is conservative (Markov) — its value is the explicit $O(1/N)$ rate with computable constants; empirically the transition occurs roughly an order of magnitude earlier (Fig. 1). Proof: supplement §S4. $\square$

Fig. 1 validates the estimator across K_4 – K_8 with per-trial random supports at several sparsities k , a full (N, ρ_2) grid with $N \in \{25, \dots, 400\}$ and $\rho_2 \in \{0.25, \dots, 2\}$, 200 Monte-Carlo trials per cell, and 95% Wilson intervals (run_second_order.py; every cell, including the derived bound (5) evaluated at that cell, is in the released JSON). Protocol notes: the headline threshold is the theorem’s rule $w_{\min}/2$ (planted weights known by design); an oracle-free *largest-gap* rule is run alongside and logged per cell in the released JSON; the full edge-domain pipeline is used, so the gradient nuisance is present and annihilated by the projection; both baselines receive *oracle* side information — the subspace method the true k

and the true $\dim \text{im } B_{2,S}$, the greedy fitter the true σ_n .

(A) NNLS is the only estimator attaining exact recovery in *every* regime — including maximal rank deficiency, e.g. rank $B_2/p = 21/56$ on K_8 — reaching probability 1 at the plotted mid- k cells by $N=400$, $\rho_2=1$ (0.99 at the densest K_8 cell, $k=28$). The oracle- k subspace baseline can lead at small N but collapses on dependent candidates (0.00 at K_8 , $k=28$): its *population* scores tie there (Remark 1(iii)), and its finite- N tie-breaking rides on sample eigen-anisotropy, a side channel the projector excitation removes entirely. Greedy stalls at sparse supports (0.42 on K_5 at $k=2$, $N=400$). (B) Recovery improves monotonically in ρ_2 uniformly across rank deficiency — the empirical face of Theorem 1(a). The bound (5) upper-bounds every cell; being a Markov bound it is conservative by one to three orders of magnitude (the empirical failure decays exponentially, the bound as $1/N$), and it is certified non-vacuously at cells where it drops to ~ 0.75 . (C) Under $\Gamma_\alpha = \rho_2[(1-\alpha)\mathbf{I} + \alpha(B_{2,S}^\top B_{2,S})^+]$ on the K_5 tetrad, with the same $w_{\min}/2$ rule (the diagonal model is deliberately misspecified for $\alpha > 0$ — that is the point), the analytic equal-image gap shrinks continuously and the empirical NNLS recovery of the specific S collapses *before* the gap reaches zero — thresholded estimation dies earlier than distinguishability — with the threshold-free endpoint at $\alpha = 1$ exactly on Theorem 1(c): identical covariances, no method can identify S . Identifiability is a property of the excitation, not the geometry alone.

5. REAL DATA: CURL-BASED VORTEX LOCALIZATION

We position the real-data study precisely: it is *vortex localization*, not filled-triangle topology recovery — the mesh below is simply connected, hence full-column-rank with no equal-image confusers, and real weather carries no latent boolean support. What it demonstrates is that the paper’s sufficient statistic localizes *genuine, unplanted* rotational structure on a real vector field. Construction: ERA5 10 m winds [21] over the Western North Pacific (Aug–Sep 2020, 0.7°/6-hourly) are subsampled to a 2.1° right-triangulated mesh (836 nodes, 2389 edges, 1554 triangles; #triangles = $|\mathcal{E}| - |\mathcal{V}| + 1$ by Euler, so B_2 is full column rank); edge flows are trapezoidal line integrals of the wind in a local equirectangular metric. By Stokes’ theorem a triangle’s curl is its circulation, so the temporally centered curl-energy score (area²-normalized to the mean-vorticity scale; no oracle parameters) ranks cyclone-bearing triangles within 4-day windows of 16 snapshots, pooled over the season’s 15 windows for evaluation. Storms translate several triangle-widths per 4-day window; centering removes

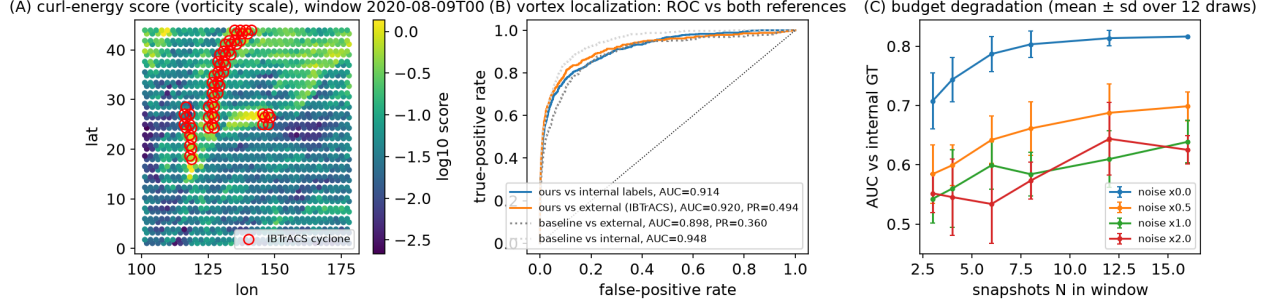


Fig. 3. Curl-based vortex localization on real data. (A) Centered curl-energy scores (vorticity scale) for one 4-day window; red circles: independent IBTrACS fixes. (B) ROC pooled over all windows: same-field vorticity reference, independent IBTrACS labels (with PR-AUC), and the classical coarse-vorticity baseline. (C) Budget degradation (mean $\pm$ sd over 12 draws; uniform centered statistic, $N \geq 3$).

the window-mean flow, so the statistic responds to the time-varying circulation a moving vortex deposits. Validation (Fig. 3B) against a *same-field consistency reference* (finite-difference vorticity on the $3\times$ -finer source grid) gives AUC 0.914 (window-bootstrap 95% CI [0.895, 0.933]), and against the *genuinely independent* IBTrACS archive [22] — 13 storms, 726 fixes — AUC 0.920 [0.873, 0.955], PR-AUC 0.494 [0.316, 0.634] at prevalence 1.8%, and $P@k = 0.483$. Against a classical pointwise-vorticity baseline at the same information budget (mesh-node winds), the edge-flow statistic is better on *all three* independent external metrics (baseline: AUC 0.898, PR-AUC 0.360, $P@k = 0.389$) and slightly lower on the internal reference (0.914 vs. 0.948), which shares the baseline’s functional. The decorrelated (GLS) variant of Prop. 1 ranks worse here (G^{-1} amplifies noise on a near-regular mesh) and is reported in the repository. FX consistency checks and planted road-network studies from earlier revisions are retained in the repository and supplement, not claimed as recovery evidence.

Limitations. The i.i.d.-Gaussian snapshot model is an idealization of windowed weather; evaluation is therefore threshold-free ranking rather than support recovery, cluster (window) bootstrap intervals quantify pooling uncertainty, and the internal reference is a same-field consistency check — only IBTrACS is independent. The vorticity-threshold sensitivity of the internal AUC (0.876–0.959 over 10^{-5} – $5 \times 10^{-5} \text{ s}^{-1}$) is reported in the released JSON. Nothing here claims a new cyclone detector; the experiment grounds the paper’s sufficient statistic on genuinely unplanted structure.

6. CONCLUSION

Identifiability of latent higher-order structure from edge flows is excitation-dependent: fully identifiable under diagonal excitation at any rank deficiency, image-only under unrestricted PSD excitation, and provably ambiguous at the projector excitation — with global analytic separation constants, a worst-case exponent equal to one isolated-triangle detection, and a consistent NNLS estimator carrying a fully explicit $O(1/N)$ bound. The trichotomy delimits both algorithmic structure learning [9, 10, 11] and subspace detection [12]; on real data the same statistic localizes unplanted cyclone circulation. Open problems: identifiability *between* the diagonal and free-PSD excitation classes (e.g. banded or Toeplitz Γ_S — where between reading off every weight and seeing only the image does the transition occur?); sharp first-order sparse thresholds over $\ker B_2$; effective sample sizes N_{eff} under temporal dependence; and tightening the Markov factor in (5) to match the empirical transition.

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Compliance with ethical standards

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Research involving human participants and/or animals. This article does not contain any studies with human participants or animals. All data used are public: ERA5 reanalysis (Copernicus/ECMWF, via the ARCO-ERA5 mirror), IBTrACS v04r01 (NOAA NCEI), TNTP road networks, and ECB reference exchange rates.

Use of AI tools (disclosure per IEEE/SPS policy). Substantial portions of the manuscript text, the source code, and the figures were generated with the assistance of an AI system (Claude, Anthropic), operating under the author’s direction across iterative adversarial-review cycles. All mathematical statements were additionally verified by the released test suite (46 tests, including exhaustive and Monte-Carlo certificates for every quoted constant), and the full revision history is preserved unaltered in the public repository. The author takes full responsibility for the content.